POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5012 | Semester V | Theory | 4 Credits

Design of RCC Structures

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5012 as Design of RCC Structures in Semester V for Civil Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public RC-design curriculum sources and is not represented as recovered official SITTTR Course 5012 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders. This handbook must not be used for construction design; use current applicable codes, approved loads/material data and qualified structural-engineer review.

> **Verified source note:** The official detailed source was inaccessible. Public sources supporting RC beam, slab, column, footing, serviceability and detailing learning are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Design of RCC Structures develops safe, code-governed design thinking for reinforced-concrete members. It links materials, loads, structural behaviour, limit states, detailing and serviceability for beams, slabs, columns and footings.

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Prerequisite foundation

Structural analysis, strength of materials, concrete technology and relevant approved design-code access.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain reinforced-concrete behaviour, materials, loads and design philosophies.
2. Analyse/design conceptual beam and slab actions using approved code procedures.
3. Explain shear, bond, development length, serviceability and detailing principles.
4. Explain column and footing behaviour and the importance of complete design checks.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain RC behaviour, limit states, loading and durable design principles.	Understand
CO2	Apply approved design sequence concepts for beams and slabs.	Apply
CO3	Explain serviceability, shear, bond and detailing requirements.	Understand
CO4	Explain the design checks required for columns and foundations.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	RC Behaviour, Materials and Design Basis	Concrete/steel behaviour, durability, loads, structural idealisation, strength and serviceability limit states.	Approx. 14 hours	CO1	Module 1
2	Beams and Slabs	Flexure, singly/doubly reinforced and T-beam concepts, one-way/two-way slabs, analysis actions and detailing intent.	Approx. 16 hours	CO2	Module 2
3	Shear, Serviceability and Detailing	Shear/diagonal tension, crack and deflection control, bond, development/splice and practical reinforcement detailing.	Approx. 16 hours	CO3	Module 3
4	Columns, Footings and Design Review	Axial/flexural column action, slenderness, footing load transfer, soil interaction and final design checks.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

RC Behaviour, Materials and Design Basis

Concrete/steel behaviour, durability, loads, structural idealisation, strength and serviceability limit states.

CO1 · Approx. 14 hours

Beams and Slabs

Flexure, singly/doubly reinforced and T-beam concepts, one-way/two-way slabs, analysis actions and detailing intent.

CO2 · Approx. 16 hours

Shear, Serviceability and Detailing

Shear/diagonal tension, crack and deflection control, bond, development/splice and practical reinforcement detailing.

CO3 · Approx. 16 hours

Columns, Footings and Design Review

Axial/flexural column action, slenderness, footing load transfer, soil interaction and final design checks.

CO4 · Approx. 14 hours

MODULE 1

RC Behaviour, Materials and Design Basis

Concrete/steel behaviour, durability, loads, structural idealisation, strength and serviceability limit states.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

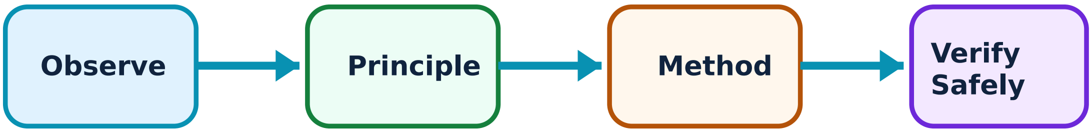
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for RC Behaviour, Materials and Design Basis: identify the system, state its principle, follow the correct method and verify the result safely.

1. Composite action and durability−

Concrete resists compression effectively while embedded steel provides tensile capacity and ductility. Cover, compaction, curing, exposure and detailing protect reinforcement and support durability.

Study and application method

Identify load path, tension/compression zones, material roles and durability inputs before any member calculation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify load path, tension/compression zones, material roles and durability inputs before any member calculation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never assume concrete strength, cover or exposure classification; use approved project data and code requirements.

2. Limit states and loads−

Design checks address ultimate strength and serviceability such as deflection, cracking and durability. Loads and combinations reflect realistic actions and uncertainty under the applicable code.

Study and application method

List permanent, imposed and relevant environmental actions; state the governing combination source and design/serviceability check required.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List permanent, imposed and relevant environmental actions; state the governing combination source and design/serviceability check required.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not mix provisions or load factors from incompatible codes/editions.

3. Design communication−

A defensible calculation identifies assumptions, code clause/version, load path, member geometry, reinforcement, checks and drawings/detailing.

Study and application method

Use a calculation template with inputs, units, code reference, sketches, result and independent reasonableness check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a calculation template with inputs, units, code reference, sketches, result and independent reasonableness check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Software output does not replace model review, code verification or professional responsibility.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Beams and Slabs

Flexure, singly/doubly reinforced and T-beam concepts, one-way/two-way slabs, analysis actions and detailing intent.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

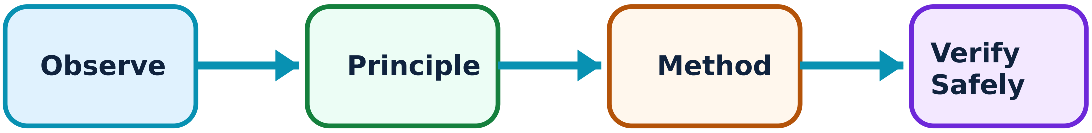
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Beams and Slabs: identify the system, state its principle, follow the correct method and verify the result safely.

1. Flexural behaviour of beams−

Bending produces compression in concrete and tension principally in steel; reinforcement ratio, section shape and material properties influence strength and ductility.

Study and application method

Determine design actions from approved analysis, identify section/cover/material inputs and follow the applicable code sequence for flexural capacity and reinforcement limits.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Determine design actions from approved analysis, identify section/cover/material inputs and follow the applicable code sequence for flexural capacity and reinforcement limits.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not apply memorised formulas without confirming units, section type, code assumptions and load effects.

2. Slab action−

One-way and two-way slabs distribute load differently depending on support and aspect ratio. Thickness, reinforcement direction, deflection and punching/shear checks can govern.

Study and application method

Establish support condition and load path, choose code-recognised analysis method and show main/distribution reinforcement intent.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Establish support condition and load path, choose code-recognised analysis method and show main/distribution reinforcement intent.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Openings, discontinuities and unusual support conditions require additional analysis and detailing.

3. T-beams and continuity−

In monolithic floor systems, slab portions may act with beams in compression. Continuous members experience positive/negative moment regions requiring correct reinforcement placement.

Study and application method

Sketch effective flange/critical moment regions according to the governing method and identify where top/bottom steel is required.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch effective flange/critical moment regions according to the governing method and identify where top/bottom steel is required.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not assume slab contribution without verifying geometry, construction and code limits.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Shear, Serviceability and Detailing

Shear/diagonal tension, crack and deflection control, bond, development/splice and practical reinforcement detailing.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

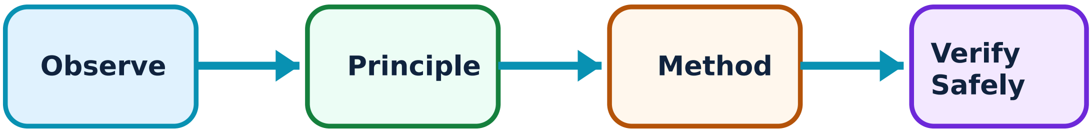
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Shear, Serviceability and Detailing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Shear and diagonal tension−

Shear can create inclined cracking and brittle failure risks; concrete capacity, shear reinforcement and detailing are checked using code-defined procedures.

Study and application method

Identify critical sections, design shear action, required capacity and reinforcement/detailing requirements under the approved code.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify critical sections, design shear action, required capacity and reinforcement/detailing requirements under the approved code.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Shear design is safety-critical; do not infer it from bending reinforcement alone.

2. Serviceability−

Deflection, cracking and vibration affect usability and durability. Span/depth checks or calculated deflection use stiffness, cracking, loading duration and code rules.

Study and application method

State serviceability criterion, loading duration and method; distinguish immediate and long-term effects where required.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State serviceability criterion, loading duration and method; distinguish immediate and long-term effects where required.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A member that meets strength may still be unacceptable in service.

3. Bond, anchorage and detailing−

Forces transfer between steel and concrete through bond. Development length, laps, hooks, confinement, spacing and cover must permit safe force transfer and constructible placement.

Study and application method

Prepare a reinforcement sketch showing bar continuation, curtailment/laps, cover and confinement based on verified code rules.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a reinforcement sketch showing bar continuation, curtailment/laps, cover and confinement based on verified code rules.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not place laps or terminate bars arbitrarily in high-force regions.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Columns, Footings and Design Review

Axial/flexural column action, slenderness, footing load transfer, soil interaction and final design checks.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

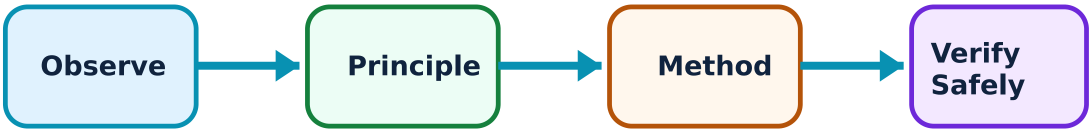
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Columns, Footings and Design Review: identify the system, state its principle, follow the correct method and verify the result safely.

1. Columns and slenderness−

Columns carry axial load often with moments from eccentricity, frame action and imperfections. Slenderness and bracing affect second-order behaviour and design procedure.

Study and application method

Identify load combination, restraint/slenderness class, section data and required axial-flexural design checks under the code.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify load combination, restraint/slenderness class, section data and required axial-flexural design checks under the code.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never treat a column as purely axial without checking eccentricity and frame effects.

2. Footings−

Footings distribute column load to soil while controlling bearing pressure, bending, one-way shear and punching shear. Soil data and load combination are fundamental inputs.

Study and application method

Use approved geotechnical allowable/design bearing data, establish footing geometry conceptually and list pressure, flexure, shear and detailing checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use approved geotechnical allowable/design bearing data, establish footing geometry conceptually and list pressure, flexure, shear and detailing checks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Foundation design requires site-specific geotechnical information; generic soil assumptions are unsafe.

3. Independent review−

Design review checks geometry, load path, units, governing combination, code version, detailing coordination and constructability before issue.

Study and application method

Perform a checklist review and obtain qualified sign-off before using any design for construction.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Perform a checklist review and obtain qualified sign-off before using any design for construction.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never issue an unreviewed learning exercise as a construction drawing.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: RC Behaviour, Materials and Design Basis

Use the concept flow to revise observation, principle, method and verification.

Module 2: Beams and Slabs

Use the concept flow to revise observation, principle, method and verification.

Module 3: Shear, Serviceability and Detailing

Use the concept flow to revise observation, principle, method and verification.

Module 4: Columns, Footings and Design Review

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a load-path and tension/compression-zone sketch for a beam/slab.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Work an instructor-approved beam or slab design example using the prescribed code.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Produce a reinforcement detailing checklist for a sample member.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare serviceability and strength checks for an assigned RC member.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — RC Behaviour, Materials and Design Basis

Explain Composite action and durability and state one correct method, application or precaution.−

Concrete resists compression effectively while embedded steel provides tensile capacity and ductility. Cover, compaction, curing, exposure and detailing protect reinforcement and support durability.

Answer framework: Identify load path, tension/compression zones, material roles and durability inputs before any member calculation.

Important point: Never assume concrete strength, cover or exposure classification; use approved project data and code requirements.

Explain Limit states and loads and state one correct method, application or precaution.−

Design checks address ultimate strength and serviceability such as deflection, cracking and durability. Loads and combinations reflect realistic actions and uncertainty under the applicable code.

Answer framework: List permanent, imposed and relevant environmental actions; state the governing combination source and design/serviceability check required.

Important point: Do not mix provisions or load factors from incompatible codes/editions.

Explain Design communication and state one correct method, application or precaution.−

A defensible calculation identifies assumptions, code clause/version, load path, member geometry, reinforcement, checks and drawings/detailing.

Answer framework: Use a calculation template with inputs, units, code reference, sketches, result and independent reasonableness check.

Important point: Software output does not replace model review, code verification or professional responsibility.

Module 2 — Beams and Slabs

Explain Flexural behaviour of beams and state one correct method, application or precaution.−

Bending produces compression in concrete and tension principally in steel; reinforcement ratio, section shape and material properties influence strength and ductility.

Answer framework: Determine design actions from approved analysis, identify section/cover/material inputs and follow the applicable code sequence for flexural capacity and reinforcement limits.

Important point: Do not apply memorised formulas without confirming units, section type, code assumptions and load effects.

Explain Slab action and state one correct method, application or precaution.−

One-way and two-way slabs distribute load differently depending on support and aspect ratio. Thickness, reinforcement direction, deflection and punching/shear checks can govern.

Answer framework: Establish support condition and load path, choose code-recognised analysis method and show main/distribution reinforcement intent.

Important point: Openings, discontinuities and unusual support conditions require additional analysis and detailing.

Explain T-beams and continuity and state one correct method, application or precaution.−

In monolithic floor systems, slab portions may act with beams in compression. Continuous members experience positive/negative moment regions requiring correct reinforcement placement.

Answer framework: Sketch effective flange/critical moment regions according to the governing method and identify where top/bottom steel is required.

Important point: Do not assume slab contribution without verifying geometry, construction and code limits.

Module 3 — Shear, Serviceability and Detailing

Explain Shear and diagonal tension and state one correct method, application or precaution.−

Shear can create inclined cracking and brittle failure risks; concrete capacity, shear reinforcement and detailing are checked using code-defined procedures.

Answer framework: Identify critical sections, design shear action, required capacity and reinforcement/detailing requirements under the approved code.

Important point: Shear design is safety-critical; do not infer it from bending reinforcement alone.

Explain Serviceability and state one correct method, application or precaution.−

Deflection, cracking and vibration affect usability and durability. Span/depth checks or calculated deflection use stiffness, cracking, loading duration and code rules.

Answer framework: State serviceability criterion, loading duration and method; distinguish immediate and long-term effects where required.

Important point: A member that meets strength may still be unacceptable in service.

Explain Bond, anchorage and detailing and state one correct method, application or precaution.−

Forces transfer between steel and concrete through bond. Development length, laps, hooks, confinement, spacing and cover must permit safe force transfer and constructible placement.

Answer framework: Prepare a reinforcement sketch showing bar continuation, curtailment/laps, cover and confinement based on verified code rules.

Important point: Do not place laps or terminate bars arbitrarily in high-force regions.

Module 4 — Columns, Footings and Design Review

Explain Columns and slenderness and state one correct method, application or precaution.−

Columns carry axial load often with moments from eccentricity, frame action and imperfections. Slenderness and bracing affect second-order behaviour and design procedure.

Answer framework: Identify load combination, restraint/slenderness class, section data and required axial-flexural design checks under the code.

Important point: Never treat a column as purely axial without checking eccentricity and frame effects.

Explain Footings and state one correct method, application or precaution.−

Footings distribute column load to soil while controlling bearing pressure, bending, one-way shear and punching shear. Soil data and load combination are fundamental inputs.

Answer framework: Use approved geotechnical allowable/design bearing data, establish footing geometry conceptually and list pressure, flexure, shear and detailing checks.

Important point: Foundation design requires site-specific geotechnical information; generic soil assumptions are unsafe.

Explain Independent review and state one correct method, application or precaution.−

Design review checks geometry, load path, units, governing combination, code version, detailing coordination and constructability before issue.

Answer framework: Perform a checklist review and obtain qualified sign-off before using any design for construction.

Important point: Never issue an unreviewed learning exercise as a construction drawing.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: RC Behaviour, Materials and Design Basis.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Beams and Slabs.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Shear, Serviceability and Detailing.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Columns, Footings and Design Review.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.

4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Concrete/steel behaviour, durability, loads, structural idealisation, strength and serviceability limit states..
2. Develop a complete answer or practical plan for: Flexure, singly/doubly reinforced and T-beam concepts, one-way/two-way slabs, analysis actions and detailing intent..
3. Develop a complete answer or practical plan for: Shear/diagonal tension, crack and deflection control, bond, development/splice and practical reinforcement detailing..
4. Develop a complete answer or practical plan for: Axial/flexural column action, slenderness, footing load transfer, soil interaction and final design checks..

LAST-MILE REVISION

Quick Revision

Module 1: RC Behaviour, Materials and Design Basis

Concrete/steel behaviour, durability, loads, structural idealisation, strength and serviceability limit states.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Beams and Slabs

Flexure, singly/doubly reinforced and T-beam concepts, one-way/two-way slabs, analysis actions and detailing intent.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Shear, Serviceability and Detailing

Shear/diagonal tension, crack and deflection control, bond, development/splice and practical reinforcement detailing.

- Match each topic to CO3.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 4: Columns, Footings and Design Review

Axial/flexural column action, slenderness, footing load transfer, soil interaction and final design checks.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Composite action and durability	Concrete resists compression effectively while embedded steel provides tensile capacity and ductility.
Limit states and loads	Design checks address ultimate strength and serviceability such as deflection, cracking and durability.
Design communication	A defensible calculation identifies assumptions, code clause/version, load path, member geometry, reinforcement, checks and drawings/detailing.
Flexural behaviour of beams	Bending produces compression in concrete and tension principally in steel; reinforcement ratio, section shape and material properties influence strength and ductility.
Slab action	One-way and two-way slabs distribute load differently depending on support and aspect ratio.
T-beams and continuity	In monolithic floor systems, slab portions may act with beams in compression.
Shear and diagonal tension	Shear can create inclined cracking and brittle failure risks; concrete capacity, shear reinforcement and detailing are checked using code-defined procedures.
Serviceability	Deflection, cracking and vibration affect usability and durability.
Bond, anchorage and detailing	Forces transfer between steel and concrete through bond.
Columns and slenderness	Columns carry axial load often with moments from eccentricity, frame action and imperfections.
Footings	Footings distribute column load to soil while controlling bearing pressure, bending, one-way shear and punching shear.
Independent review	Design review checks geometry, load path, units, governing combination, code version, detailing coordination and constructability before issue.

LEARNING RESOURCES

References

Recommended RCC-design references

1. A. K. Jain, Reinforced Concrete: Limit State Design.
2. N. Krishna Raju, Design of Reinforced Concrete Structures.
3. Current applicable reinforced-concrete design code and commentary.

Approved public RC-design sources

- https://pec.ac.in/sites/default/files/pdf/civil_structures_syllabus_0.pdf

- https://neurodiversity-engineering.media.uconn.edu/wp-content/uploads/sites/3154/2023/05/CE-3640-Fall-2022-Syllabus-new.pdf

These sources support the study handbook while official Course 5012 content is unavailable; they do not replace current applicable codes or qualified structural design.